# DAY 4 | DOMAIN JOIN & GROUP POLICY VERIFICATION

Windows Server 2025 Active Directory Lab | 25 October 2026

**Objective:** Integrate a Windows 11 client with the existing Active Directory domain, validate authentication and DNS service discovery, confirm user-targeted Group Policy application, and test enforcement of a configured restriction.

## Lab Environment

| Component | Verified configuration |
|---|---|
| Domain / DNS namespace | adlab.test |
| Domain controller | DC01.adlab.test — Windows Server 2025 |
| Domain client | CLIENT01 — Windows 11 Pro |
| Domain account | ADLAB\jsmith (John Smith) |
| Organizational unit | adlab.test/Lab-Users |
| Applied GPO | Lab-User security policy |
| Virtualization | Oracle VirtualBox |

## Implementation & Troubleshooting

• Installed Windows 11 Pro on CLIENT01 and configured the client on the VirtualBox ADLab-NAT network.

• Configured CLIENT01 to use domain controller DNS at 10.0.2.3 rather than the external/default resolver.

• Troubleshot an initial failed LDAP SRV lookup. After flushing the DNS resolver cache, the client successfully resolved **_ldap._tcp.dc._msdcs.adlab.test** to **dc01.adlab.test** on port **389**.

• Confirmed domain controller discovery with **nltest /dsgetdc:adlab.test**, which returned DC01 at 10.0.2.3.

• Joined CLIENT01 to the **adlab.test** domain and received the Windows welcome-to-domain confirmation.

• Signed in to CLIENT01 as the domain user John Smith and verified the security context using **whoami**.

## Verification Evidence

| Check / command | Observed outcome |
|---|---|
| whoami | adlab\jsmith |
| nslookup -type=SRV _ldap._tcp.dc._msdcs.adlab.test | DC01 advertised as LDAP service endpoint (port 389) |
| nltest /dsgetdc:adlab.test | DC01.adlab.test discovered; command completed successfully |
| gpresult /r | Lab-User security policy listed under Applied Group Policy Objects |

| Check / command | Observed outcome |
| --- | --- |
| gpresult /h ... | HTML report: user ADLAB\jsmith in Lab-Users; Group Policy Infrastructure and Registry reported Success |
| Administrative Templates → System | Prevent access to the command prompt = Enabled; winning GPO = Lab-User security policy |
| Interactive cmd test | “The command prompt has been disabled by your administrator.” |

## Snapshot & Recovery Checkpoint

Created Day 4 VirtualBox snapshots for **CLIENT01** and **DC01** after orderly shutdown. DC01’s earlier Day 1–Day 3 snapshots remain visible in its snapshot tree. These snapshots provide lab rollback points; they are not substitutes for independent backups.

## Outcome & Skills Demonstrated

The client authenticated to Active Directory, discovered the domain controller through DNS, received the linked user Group Policy, and visibly enforced the configured Command Prompt restriction. This lab demonstrates Windows client deployment, AD DS domain joining, DNS troubleshooting, domain authentication, GPO result analysis, endpoint policy validation, and VM lifecycle management.

## Suggested GitHub Evidence Files

Capture or upload: domain-join confirmation; **whoami** output; successful SRV lookup and **nltest** results; **gpresult /r**; the HTML GPO report showing the enabled policy; the Command Prompt blocked message; and both Day 4 snapshot trees. Review screenshots for credentials or sensitive identifiers before publishing.

*Documentation scope: Results are based on observed screenshots and commands from this lab session; no unverified settings or additional tests are claimed.*